

MARTIN ANDREEV ASENOV

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EDUCATION

The University of Edinburgh and Heriot-Watt University Sep 2017 - Jan 2022
Ph.D. in Robotics and Autonomous Systems

- “Architectures for Online Simulation-Based Inference Applied to Robot Motion Planning”
- Supervised by Prof. Subramanian Ramamoorthy and Dr. Kartic Subr

The University of Edinburgh and Heriot-Watt University Sep 2016 - Aug 2017
MRes in Robotics and Autonomous Systems

- “Active Sensing of Spatiotemporal Phenomena with a UAV”
- Supervised by Prof. Subramanian Ramamoorthy

The University of Edinburgh Sep 2012 - May 2016
BSc (Hons) Artificial Intelligence and Computer Science

- “Dynamic Model of Interactions Between Orientation Selective Neurons in Primary Visual Cortex”
- Supervised by Dr. Mark van Rossum
- International Exchange Student at The University of California, Irvine (Sep 2014 - Jun 2015)

WORK EXPERIENCE

Parexel AI Labs Sep 2025 - Current
Senior Machine Learning Engineer I *Edinburgh, UK (Remote)*

- Framed document generation as RL with verifiable rewards; built a token-efficient OOXML library beating OpenAI/Anthropic document skills at ~half the tokens and 40% higher reward
- Corpus-level QA over clinical-trial documents, used internally by 20K+ employees

Huawei Technologies Research & Development Aug 2022 - Sep 2025
Senior Machine Learning Engineer *Edinburgh, UK*

- Shipped a time-series framework for serverless resource prediction, saving thousands of £/month
- Co-developed DAM, a transformer foundation model for zero-shot time-series forecasting
- Led a 7-person team on zero-shot runtime prediction across GPUs, NPUs and CPUs (6–10% error)
- Published 9 papers (ICLR, ICML, EuroSys, SoCC, EuroMLSys, IC2E) and 2 patents in 3 years

Hyperscience Apr 2022 - Jul 2022
Machine Learning Engineer *Edinburgh, UK (Remote)*

- Improved document rotation-correction accuracy 15% via domain adaptation and reduced preprocessing

The University of Edinburgh Apr 2021 - Mar 2022
Research Associate *Edinburgh, UK*

- Breast cancer classification based on combined pCLE images and Raman spectroscopy
- Adaptive scanning strategies for ultrasound examinations

Amazon, PrimeAir Oct 2018 - Dec 2018
Applied Scientist Intern *Graz, Austria*

- Image segmentation from limited data using domain adaptation and gradient reversal layer

The University of Edinburgh
Research Assistant

Jan 2017 - Jun 2017; Nov 2017 - Sep 2018
Edinburgh, UK

- Collaboration between DSTL, Heriot-Watt University, BMT and REDALERT
- Data-driven fluid simulation approach to characterize gas dynamics

Thales Avionics
Software Integration and Characterization Developer

Jul 2015 - Aug 2015
Irvine, California, USA

- Extended Google's Blockly visual programming editor to a test automation tool

SELECTED PUBLICATIONS

Retrieval or Representation? Reassessing Benchmark Gaps in Multilingual and Visually Rich RAG, M. Asenov, K. Benkirane, D. Goldwater, A. Ghodsi, I Can't Believe It's Not Better: Where Large Language Models Need to Improve, **ICLR '26**

DISCO: Document Intelligence Suite for COmparative Evaluation, K. Benkirane, D. Goldwater, M. Asenov, A. Ghodsi, Multimodal Intelligence: Next Token Prediction & Beyond, **ICLR '26**

Higher Acceptance Rates for Speculative Decoding with Randomised Drafting, W. Toner, M. Asenov, R. Singh, A. Joosen, Tiny Titans: The next wave of On-Device Learning for Foundational Models (TTODLer-FM), **ICML '25**

Lightweight Online Adaption for Time Series Foundation Model Forecasts, T. L. Lee, W. Toner, R. Singh, A. Joosen, M. Asenov, **ICML '25**

Serverless Cold Starts and Where to Find Them, A. Joosen, A. Hassan, M. Asenov, R. Singh, L. Darlow, J. Wang, Q. Deng, A. Barker, **EuroSys '25**

Performance of Zero-Shot Time Series Foundation Models on Cloud Data, W. Toner, T. L. Lee, A. Joosen, R. Singh, M. Asenov, I Can't Believe It's Not Better: Challenges in Applied Deep Learning Workshop, **ICLR '25**

DAM: Towards a Foundation Model for Forecasting, L. N. Darlow, Q. Deng, A. Hassan, M. Asenov, R. Singh, A. Joosen, A. Barker, A. Storkey, **ICLR '24**

FoldFormer: Sequence Folding and Seasonal Attention for Fine-grained Long-term FaaS Forecasting, L. N. Darlow, A. Joosen, M. Asenov, Q. Deng, J. Wang, A. Barker, **EuroMLSys '23**

Learning to Score: Tuning Cluster Schedulers through Reinforcement Learning, M. Asenov, Q. Deng, G. Yeung, A. Barker, **IC2E '23**

Architectures for online simulation-based inference applied to robot motion planning, M. Asenov, **PhD Thesis**

Vision-Based System Identification and 3D Keypoint Discovery using Dynamics Constraints, M. Jaques, M. Asenov, M. Burke, T. Hospedales, Workshop on Physical Reasoning and Inductive Biases for the Real World, **NeurIPS '21**

Inversion of Ultrafast X-ray Scattering with Dynamics Constraints, M. Asenov, N. Zotev, S. Ramamoorthy, A. Kirrander, Workshop on Machine Learning and the Physical Sciences, **NeurIPS '20**

Vid2Param: Modelling of Dynamics Parameters from Video, M. Asenov, M. Burke, D. Angelov, T. Davchev, K. Subr, S. Ramamoorthy, **RAL**, Vol 5(2): 414-421. Presented at **ICRA '20**

Active Localization of Gas Leaks using Fluid Simulation, M. Asenov, M. Rutkauskas, D.T. Reid, K. Subr, S. Ramamoorthy, **RAL**, Vol 4(2): 1776-1783. Presented at **ICRA '19**